

Samir Yunus

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[Portfolio](#)

EDUCATION

Rutgers University

Bachelor of Science, Aerospace Engineering | GPA 3.7 | Dean's List

Brookdale Community College

Associates of Science, Mechanical Engineering | GPA 3.5 | Dean's List

New Brunswick, NJ

September 2024 - May 2027

Lincroft, NJ

September 2022 - May 2024

PROFESSIONAL EXPERIENCE

Rutgers Rocket Propulsion Lab (RRPL)

Structures Team Member

Piscataway, NJ

September 2025 – Expected May 2027

- Co-led development of RRPL's first deployable air-brake system targeting 10,000 ft apogee for IREC competition
- Performed CFD analysis in ANSYS Fluent (Mach 0.2–0.9) to evaluate drag and airflow characteristics; validated simulation results through wind tunnel testing
- Conducted structural FEA in ANSYS Mechanical reducing prototype redesign iterations prior to manufacturing
- Designed and fabricated ignition sled hardware supporting avionics integration for a 100,000 ft competition rocket

Rutgers University Department of Mechanical & Aerospace Engineering

Automated Rust Detection Researcher

Piscataway, NJ

September 2024 - December 2024

- Developed a computer vision pipeline in Python and semantic segmentation CNNs for corrosion detection, achieving ~90% detection accuracy across 300+ labeled samples
- Processed and labeled 300+ image samples to expand training data and improve segmentation performance
- Authored onboarding documentation for deploying Label Studio in WSL environments

PROJECTS

Reusable Rocket Launch & Landing Simulation

MATLAB, Simulink

- Developed a 6-DOF reusable launch vehicle simulation modeling ascent, boost-back, atmospheric entry, and landing trajectories in MATLAB/Simulink
- Implemented PID-based guidance and trajectory propagation with atmospheric drag modeling
- Integrated real-time FlightGear visualization through UDP FDM interface

Thrust Vector Control (TVC) Prototype

SolidWorks, Arduino, Controls

- Co-led a 5-member team developing a thrust-vectoring gimbal system for small-scale rocket stabilization and control experiments
- Designed and fabricated a custom 3D-printed thrust-vectoring gimbal in SolidWorks integrating servo actuation and inertial sensing
- Implemented closed-loop attitude control using MPU6050 IMU feedback, achieving stabilization within 1–2°

Toyota Chaser Manual Transmission Swap

CAD, CNC Machining, Mechanical Integration

- Executed complete automatic-to-manual conversion including CAD design and custom component fabrication
- Designed and CNC-machined hydraulic spacer components restoring OEM drivability

TECHNICAL SKILLS

Software: SolidWorks, ANSYS Fluent, ANSYS Mechanical, MATLAB/Simulink, Python, LabVIEW

Manufacturing: CNC, Composites, 3D printing, Instrumentation, Oscilloscopes, Multimeters, Mechanical Assembly

Analysis: CFD, FEA, Structural Analysis, Controls

ADDITIONAL EXPERIENCE

Visual Computer Solutions - Jobs4Blue

Senior Account Representative

Freehold, NJ

April 2023 – Present

- Managed communications across 160+ client organizations and maintained operational workflows
- Coordinated scheduling and task allocation across a 50-person team while tracking 200+ daily activities

PUBLIC PRESENTATIONS

Guest Speaker

The Mythology and Science of Eclipses

- Delivered a 120-minute lecture on eclipse science and astronomy to 50+ attendees

AFFILIATIONS & ADDITIONAL

Rutgers Car Club - Personal automotive project featured in Rutgers Magazine for engineering innovation and craftsmanship

Active AIAA Member - Pursuing High-Power Rocketry Level 1 & 2 Certification